Practical - 7

Problem Statements -

Q1. Regression with T-test: Using the teachers rating data set, does gender affect teaching evaluation rates?

Q2. Regression with ANOVA: Using the teachers' rating data set, does beauty score for instructors differ by age?

Q3. Correlation: Using the teachers' rating dataset, Is teaching evaluation score correlated with beauty score?

Solution- Here is the complete code

Conclusion - The analysis of the TeachingRatings dataset using regression, t-test, ANOVA, and correlation shows that gender, age group, and beauty score have statistically significant relationships with teaching evaluation scores or beauty scores. The t-test in Q1 indicates a significant difference in teaching evaluation scores between male and female instructors. The ANOVA in Q2 shows a significant difference in instructors' beauty scores across the different age groups. The correlation/regression analysis in Q3 shows a statistically significant positive relationship between beauty score and teaching evaluation score. Therefore, the statistical tests provide evidence of significant relationships among the variables examined in this dataset.

```
# Import required libraries

import pandas as pd
import numpy as np

import statsmodels.api as sm
import statsmodels.formula.api as smf
from scipy import stats

import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings('ignore')
```

```
# Load TeachingRatings dataset

url = "https://raw.githubusercontent.com/vincentarelbundock/Rdatasets/master/csv/AER/TeachingRatings.csv"

df = pd.read_csv(url)

# Display first 5 rows
df.head()
```

	rownames	minority	age	gender	credits	beauty	eval	division	native	tenure	students	allstudents	prof
0	1	yes	36	female	more	0.289916	4.3	upper	yes	yes	24	43	1
1	2	no	59	male	more	-0.737732	4.5	upper	yes	yes	17	20	2
2	3	no	51	male	more	-0.571984	3.7	upper	yes	yes	55	55	3
3	4	no	40	female	more	-0.677963	4.3	upper	yes	yes	40	46	4
4	5	no	31	female	more	1.509794	4.4	upper	yes	yes	42	48	5

Next steps: Generate code with df | New interactive sheet

```
# Check number of rows and columns

print("Shape of dataset:", df.shape)

# Display column names
print("\nColumns:")
print(df.columns.tolist())
```

```
Shape of dataset: (463, 13)

Columns:
['rownames', 'minority', 'age', 'gender', 'credits', 'beauty', 'eval', 'division', 'native', 'tenure', 'students', 'allstudents', 'prof']
```

```
# Create dummy variable for female
```

```
df['female'] = (df['gender'] == 'female').astype(int)

# Check
df[['gender', 'female']].head(10)
```

	gender	female
0	female	1
1	male	0
2	male	0
3	female	1
4	female	1
5	male	0
6	female	1
7	female	1
8	female	1
9	male	0

Q1 Regression with T-test: Using the teachers rating data set, does gender affect teaching evaluation rates?

```
# Regression model

model_q1 = smf.ols('eval ~ female', data=df).fit()

# Display regression results
print(model_q1.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   eval   R-squared:                       0.022
Model:                            OLS   Adj. R-squared:                  0.020
Method:                 Least Squares   F-statistic:                     10.56
Date:                Sun, 27 Sep 2026   Prob (F-statistic):            0.00124
Time:                        09:35:41   Log-Likelihood:                -378.50
No. Observations:                 463   AIC:                             761.0
Df Residuals:                     461   BIC:                             769.3
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      4.0690      0.034    121.288      0.000       4.003       4.135
female        -0.1680      0.052     -3.250      0.001      -0.270      -0.066
==============================================================================
Omnibus:                       17.625   Durbin-Watson:                   1.368
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               18.970
Skew:                          -0.496   Prob(JB):                     7.60e-05
Kurtosis:                       2.981   Cond. No.                         2.47
==============================================================================
```

```
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Result: The p-value for female is 0.001, which is less than 0.05. Therefore, the null hypothesis is rejected. Hence, there is a statistically significant difference in teaching evaluation scores between male and female instructors.

Ques 2. Using the teachers' rating data set, does beauty score for instructors differ by age?

```
# Create age groups

df['age_group'] = np.where(
    df['age'] <= 40,
    '40 years and younger',
    np.where(
        df['age'] < 57,
        'between 40 and 57 years',
        '57 years and older'
    )
)

# Check the groups
print(df['age_group'].value_counts())
```

```
age_group
between 40 and 57 years    228
57 years and older         122
40 years and younger       113
Name: count, dtype: int64
```

```
# Regression model

model_q2 = smf.ols(
    'beauty ~ C(age_group)',
    data=df
).fit()

print(model_q2.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 beauty   R-squared:                       0.071
Model:                            OLS   Adj. R-squared:                  0.067
Method:                 Least Squares   F-statistic:                     17.60
Date:                Sun, 27 Sep 2026   Prob (F-statistic):           4.32e-08
Time:                        09:41:24   Log-Likelihood:                -529.47
No. Observations:                 463   AIC:                             1065.
Df Residuals:                     460   BIC:                             1077.
Df Model:                           2
Covariance Type:            nonrobust
================================================================================================================
                                               coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------------------------
Intercept                                    0.3362      0.072      4.692      0.000       0.195       0.477
C(age_group)[T.57 years and older]          -0.5820      0.099     -5.852      0.000      -0.777      -0.387
C(age_group)[T.between 40 and 57 years]     -0.3713      0.088     -4.237      0.000      -0.544      -0.199
```

```
==============================================================================
Omnibus:                       11.586   Durbin-Watson:                   0.832
Prob(Omnibus):                  0.003   Jarque-Bera (JB):               12.114
Skew:                           0.394   Prob(JB):                      0.00234
Kurtosis:                       2.913   Cond. No.                         4.41
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

```
# Generate ANOVA table

anova_q2 = sm.stats.anova_lm(
    model_q2,
    typ=2
)

# Rename row to match the practical
anova_q2 = anova_q2.rename(
    index={'C(age_group)': 'age_group'}
)

print(anova_q2)
```

```
                sum_sq     df          F        PR(>F)
age_group    20.422743    2.0  17.597558  4.322550e-08
Residual    266.925153  460.0        NaN           NaN
```

Interpretation: Since the p-value (4.322549×10^{-8}) is less than 0.05, we reject the null hypothesis. Hence, there is a significant difference in instructors' beauty scores across the different age groups.

Q3. Correlation: Using the teachers' rating dataset, Is teaching evaluation score correlated with beauty score?

```
# OLS regression
model_q3 = smf.ols(
    'eval ~ beauty',
    data=df
).fit()

print(model_q3.summary())

# Pearson correlation
correlation, p_value = stats.pearsonr(
    df['beauty'],
    df['eval']
)

print("\nPearson Correlation =", correlation)
print("P-value =", p_value)
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   eval   R-squared:                       0.036
Model:                            OLS   Adj. R-squared:                  0.034
Method:                 Least Squares   F-statistic:                     17.08
```

```
Date:                Sun, 27 Sep 2026   Prob (F-statistic):           4.25e-05
Time:                        09:45:46   Log-Likelihood:                -375.32
No. Observations:                 463   AIC:                             754.6
Df Residuals:                     461   BIC:                             762.9
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      3.9983      0.025    157.727      0.000       3.948       4.048
beauty         0.1330      0.032      4.133      0.000       0.070       0.196
==============================================================================
Omnibus:                       15.399   Durbin-Watson:                   1.410
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               16.405
Skew:                          -0.453   Prob(JB):                     0.000274
Kurtosis:                       2.831   Cond. No.                         1.27
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Pearson Correlation = 0.18903909062278274
P-value = 4.247115507786e-05
```

Interpretation: Since the p-value for beauty is less than 0.05, we reject the null hypothesis. Therefore, there is a statistically significant positive relationship between beauty score and teaching evaluation score. The coefficient of beauty is 0.1330, indicating that a one-unit increase in beauty score is associated with an increase of approximately 0.133 in the teaching evaluation score. The (R^2) value of 0.036 indicates that beauty explains about 3.6% of the variation in evaluation scores.

```
Start coding or generate with AI.
```